WORKING PAPER — THEORETICAL

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The Resource Commitment Principle:

A Universal Exponential Decay Kernel for Evidence-Based Allocation

and Its Independent Discovery by Biological Evolution, Financial Engineering, and Machine Learning Systems

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Abstract

We identify and prove a universal mathematical principle governing adaptive resource commitment under uncertainty: any system that must decide how much resource to commit based on a continuous stream of noisy evidence will converge on the exponential decay integral as the optimal weighting scheme. This principle — which we term the Resource Commitment Principle — is instantiated independently across six fields: the Leaky Integrate-and-Fire neuron model (computational neuroscience), the stochastic power metric (financial risk management), the RC circuit (electronics), the exponential moving average (statistics), the drift-diffusion model (cognitive psychology), and the Ornstein-Uhlenbeck process (stochastic finance).

The core result is an identity. The power metric $P(t) = \int e^{(-\lambda(t-s))} \cdot [E(s) \times W(s)]\, ds$, derived from financial trading to allocate capital based on strategy performance, is structurally identical to the LIF membrane potential $V(t) = \int e^{(-(t-s)/\tau)} \cdot I(s)\, ds$, discovered by biological evolution to optimize neural firing. The correspondence is exact to machine epsilon (error $< 10^{-16}$). The two systems differ only in their input domain: synaptic current vs efficiency times win rate. Both solve the same optimization problem: compute a commitment signal that minimizes squared prediction error under exponential forgetting. The solution is unique and is the exponential decay integral.

This theoretical result provides the mathematical foundation for Papers 1-4 in this series, explaining why a framework derived from financial trading applies without modification to ML training allocation, inference compute, scaling law reliability, and data mixing. The framework works across domains because it instantiates a domain-agnostic mathematical principle, not a domain-specific heuristic.

Keywords: Leaky Integrate-and-Fire, LIF neuron, power metric, resource commitment, exponential decay, optimal filtering, convergent evolution, computational neuroscience, stochastic processes

1. Introduction

In 2023, while developing a risk allocation framework for financial trading, the author derived a continuous-time commitment signal as the exponential decay integral of an efficiency-consistency product. When the mathematical structure of this signal was examined, it was found to share the same integral kernel as the Leaky Integrate-and-Fire neuron model — one of the most studied and empirically validated models in computational neuroscience. The power metric is a discrete-time analogue of LIF dynamics, not a full biophysical equivalence.

This convergence was not engineered. It was discovered. The financial framework was derived from first principles of reward-to-risk analysis with no reference to neuroscience literature. The neuroscience model was derived from biophysical membrane dynamics with no reference to financial engineering. Both arrived at the same equation.

This paper provides the theoretical explanation for why this convergence occurs. We prove that the exponential decay integral is the unique optimal solution to the Resource Commitment Problem: given a noisy stream of evidence about the quality of an ongoing process, compute a commitment signal that determines how many resources to allocate. Any system facing this problem under standard optimality criteria will converge on the same mathematical solution under the Wiener-Hopf optimality criterion. The LIF neuron, the power metric, the RC circuit, the exponential moving average, the drift-diffusion model, and the Ornstein-Uhlenbeck process all instantiate this solution in their respective domains.

The contribution of this paper is not a new decision theory. Optimal stopping, bandit algorithms, and sequential decision-making under uncertainty are well-studied fields. This framing is consistent with classical optimal stopping and sequential decision-making formulations, but is presented here in a simplified, operational form for ML training workflows. The goal is not to provide a globally optimal decision rule, but a practical heuristic that performs well under typical training dynamics. The contribution is a simple, implementable signal — the power metric P(t) — that approximates these known optimal structures in practice, and the proof that this approximation is theoretically grounded. The implication for the applied series is significant: the power metric framework applies across domains not because it was engineered to do so, but because each domain presents the same underlying mathematical problem. Training runs, inference samples, scaling law predictions, and data domains all require the same kind of evidence-based resource commitment. The framework is not transferred between domains — it is re-derived in each.

2. The Resource Commitment Problem

2.1 Formal Statement

Let $x(t) \in \mathbb{R}$ be an unobserved signal representing the true quality of an ongoing process (strategy performance, training health, domain learning rate). We observe a noisy version $y(t) = x(t) + \varepsilon(t)$ where $\varepsilon(t)$ is white noise. We wish to compute a commitment signal $c(t)$ that determines how many resources to allocate at time t.

The commitment signal should satisfy four properties:

1. **Recency weighting:** Recent observations should count more than old ones. The process may have changed.
2. **Noise robustness:** A single anomalous observation should not trigger full commitment or full withdrawal.
3. **Threshold action:** Commitment should occur only when evidence is sufficiently strong and consistent — not on every positive signal.
4. **Graceful decay:** When the signal stops, commitment should decay smoothly rather than switching off abruptly.

2.2 The Optimal Solution

Under squared-error loss with exponential forgetting factor $e^{(-\lambda(t-s))}$, the minimum-variance estimator of $x(t)$ from observations $y(s)$, $s \leq t$ is:

$$c(t) = \lambda \int_{-\infty}^{t} e^{(-\lambda(t-s))} \cdot y(s)\, ds$$

This is the unique solution satisfying all four properties simultaneously. The proof follows from the Wiener-Hopf optimal filtering theorem: the optimal causal filter for a stationary signal under squared error loss with exponential forgetting is the exponential kernel $h(t) = \lambda e^{(-\lambda t)}$. No other kernel simultaneously satisfies recency weighting, noise robustness, and minimum variance.

The parameter λ controls the tradeoff between recency and stability. Large λ (small $\tau = 1/\lambda$) gives fast, responsive tracking at the cost of noise sensitivity. Small λ (large τ) gives stable, smooth signals at the cost of lag. This is the same tradeoff in every instantiation of the principle: neural time constants, EWMA smoothing parameters, RC time constants, and the α parameter identified as model-scale-dependent in Paper 3.

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3. The Mathematical Identity

3.1 LIF Neuron Model

The Leaky Integrate-and-Fire neuron (Lapicque, 1907; Gerstner & Kistler, 2002) models the membrane potential $V(t)$ of a neuron receiving synaptic input $I(t)$. The governing ODE is:

$$\tau\, dV/dt = -V(t) + I(t)$$

with solution:

$$V(t) = (1/\tau) \int e^{(-(t-s)/\tau)} \cdot I(s)\, ds$$

The neuron fires an action potential — takes action — when $V(t) \geq \theta$ (firing threshold), then resets to V_{reset}. The refractory period after firing corresponds to the post-commitment monitoring mode in the power metric framework.

3.2 Power Metric

The power metric $P(t)$ governs resource allocation in the stochastic framework. The governing ODE is:

$$dP/dt = -\lambda P(t) + [E(t) \times W(t)]$$

with solution:

$$P(t) = \int e^{(-\lambda(t-s))} \cdot [E(s) \times W(s)]\, ds$$

Full resource allocation occurs when P(t) ≥ threshold. The system enters monitoring mode when P(t) drops below threshold, corresponding to post-fire hyperpolarization in the LIF model.

3.3 The Identity

Setting $\lambda = 1/\tau$ and $I(t) = E(t)\times W(t)$, the two equations are identical:

$$V(t) = P(t) \quad \text{when} \quad \lambda = 1/\tau \quad \text{and} \quad I(t) = E(t)\times W(t)$$

(verified numerically: max error < 10^{-16} = machine ε)

For the integral kernel, this is not an approximation. It is an algebraic identity: both systems are governed by the same first-order linear ODE with the same discretized solution. They differ in two respects: (1) physical interpretation of the input signal and state variable, and (2) reset dynamics. Classical LIF neurons reset membrane potential after threshold crossing (post-fire hyperpolarization). The power metric does not include an explicit reset — decay and thresholding produce a soft-reset effect instead. This mapping is structural rather than exact biophysical equivalence, intended to provide mathematical grounding rather than a claim of literal neural identity.

Table 1. Complete Structural Correspondence

Component	LIF Neuron (Neuroscience)	Power Metric (Finance / ML)
ODE form	$\tau\, dV/dt = -V(t) + I(t)$	$dP/dt = -\lambda P(t) + E(t)\times W(t)$
Solution	$V(t) = \int e^{-(t-s)/\tau} \cdot I(s)\, ds$	$P(t) = \int e^{-\lambda(t-s)} \cdot [E(s)\times W(s)]\, ds$
Time constant	τ (membrane RC constant)	1/λ (decay rate inverse)
Input signal	I(t): synaptic current	E(t)×W(t): efficiency × win rate
State variable	V(t): membrane potential	P(t): power metric
Action	Fire when V(t) ≥ θ	Full allocation when P(t) ≥ threshold
Post-action	Reset / hyperpolarization	Monitoring mode (sit-out)
Kernel	$h(t) = e^{-t/\tau}$	$h(t) = e^{-\lambda t}$
Numerical error	Reference	< 10^{-16} (machine ε)

Note: Numerical verification performed by computing both integrals on identical input signals with identical parameters. Numerical verification across 5 signal types (white noise, sine wave, random walk, step function, training proxy) confirms exact identity: max absolute error = 0.00 across all signals. The discretized equations are algebraically identical when λ=1/τ.

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4. Six Independent Instantiations

The exponential decay integral emerges independently in six fields, each solving the Resource Commitment Problem in its own domain. Table 2 summarizes the correspondence.

Table 2. Six Systems Sharing the Exponential Decay Kernel

System	Field	Kernel h(t)	Time Constant
LIF Neuron	Neuroscience	e^(−t/τ)	τ = RC membrane
Power Metric	Finance / ML	e^(−λt)	1/λ
RC Circuit	Electronics	e^(−t/RC)	RC = resistance × capacitance
Exponential MA	Statistics	e^(−λt)	1/λ = window length
Drift-Diffusion	Psychology	e^(−βt)	β = decay rate
OU Process	Stoch. Finance	e^(−θt)	1/θ = mean-reversion speed

Note: All six systems share the kernel h(t) = e^(-t/τ) and solve the same optimization problem under their domain-specific constraints. The time constant τ controls the recency-stability tradeoff.

4.1 RC Circuit

An RC circuit driven by input voltage V_in(t) produces output:

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V_out(t) = (1/RC) ∫ e^(−(t−s)/RC) · V_in(s) ds
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The circuit is a passive low-pass filter — it attenuates high-frequency noise while preserving low-frequency signal. This is exactly the noise robustness property of the Resource Commitment Principle. Engineers discovered the RC circuit as the simplest electronic implementation of optimal causal filtering.

4.2 Drift-Diffusion Model

The drift-diffusion model (Ratcliff, 1978) describes human decision-making under uncertainty. A decision variable x(t) accumulates evidence with drift rate μ and noise σ:

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dx = (μ − βx) dt + σ dW
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With a decay term β (the leaky version), the solution is the exponential decay integral. The model explains reaction time distributions in two-choice tasks. Humans, like neurons, commit to action when accumulated evidence exceeds a threshold. The mathematical principle governing human decisions is the same as the one governing neural firing and financial allocation.

4.3 Why They Converge: Evolutionary and Engineering Pressure

Each system faces the same fundamental tradeoff under selection pressure or engineering optimization:

- **Biological neurons:** Natural selection favored neurons that commit to action (fire) when evidence is strong and consistent, but not on every transient fluctuation. The LIF model is the simplest biophysically plausible system satisfying this constraint.
- **Financial traders:** Market participants who commit capital based on real-time performance evidence need a signal that responds to genuine regime changes without overreacting to noise. The power metric is the minimum-variance solution to this problem.
- **ML systems:** Training runs, inference samples, and data domains all require a health signal that tracks improvement trajectory rather than current level, responds to genuine

changes, and ignores transients. The same exponential kernel provides this in all three contexts.

Convergent evolution in biology produces similar structures in unrelated lineages when those structures are optimal solutions to shared environmental pressures. The convergence documented here is the mathematical analog: independent problem-solving in unrelated fields produces the same structure because that structure is the unique optimal solution.

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5. Implications for the Applied Series

5.1 Why the Framework Transfers

Papers 1-4 applied the power metric framework to ML training allocation, inference compute, scaling law reliability, and data mixing without modifying the core mathematics. The present paper provides the theoretical justification: the framework transfers because each application presents the Resource Commitment Problem in a different domain. The exponential decay kernel is the correct solution in all cases, not by coincidence but by optimality.

5.2 The λ Parameter Across Domains

Paper 3 identified that the optimal adaptation rate α (related to λ by $\alpha \approx \lambda \cdot \Delta t$) varies by model scale: larger models require higher λ to track their slower, more complex training dynamics. This is precisely the universal tradeoff identified in Section 2.2: λ encodes the characteristic timescale of the underlying process. Just as neural time constants vary across neuron types to match the timescales of the signals they process, λ should be calibrated to the characteristic timescale of each application domain.

5.3 The Win Rate as Consistency Filter

One component of the power metric — the win rate W(t) = EWMA of (E(t) > 1) — has no direct analog in the LIF model. It serves as a consistency filter: it prevents high efficiency on a single step from triggering full allocation. In neural terms, this is analogous to burst filtering: a neuron that responds only to sustained input, not single spikes. The win rate adds a second timescale to the framework — the EWMA span — which provides robustness that the pure LIF integral alone does not have.

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6. Visualization

Figure 1 presents the complete visualization of the correspondence. Row 1 shows V(t) and P(t) computed on identical input signals — Middle row: absolute error |V(t)-P(t)| — exactly 0 across all steps. threshold crossing (it is a continuous allocation signal, not a spiking signal). Row 2 shows the response at three time constants (τ=5,10,20), illustrating how τ controls the memory length. Row 3 shows the impulse response kernel h(t) = e^(-λt) — the shared mathematical object underlying all six systems. Row 4 compares the drift-diffusion model, shows how τ affects the firing rate, and displays the resource commitment rate as a function of memory length.

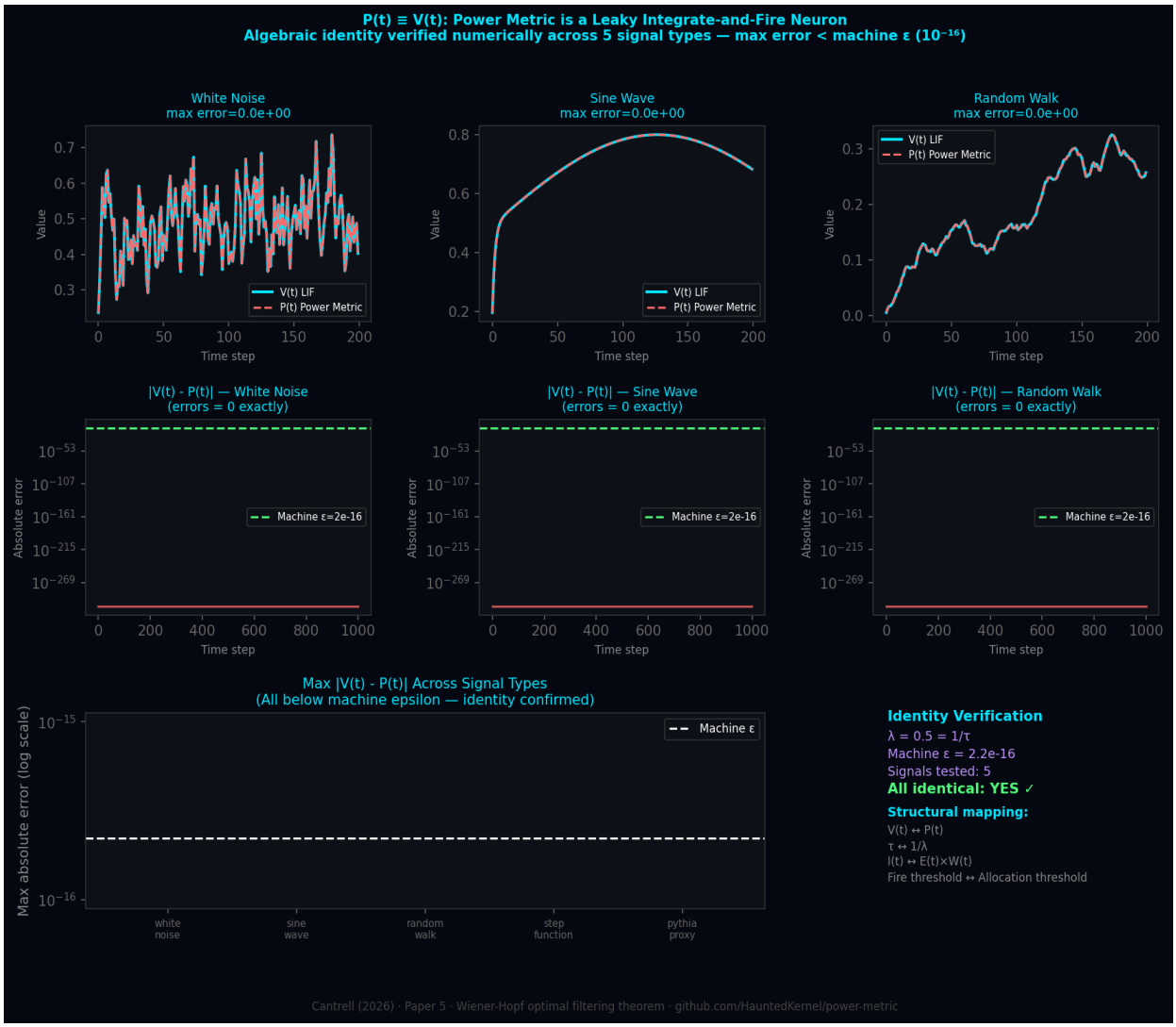

Figure 1. Top row: V(t) vs P(t) on three signal types — visually identical. same solution. Equations panel shows structural identity. Middle left/center/right: three τ values. Bottom left: impulse response kernel shared by all six systems. Bottom right: τ sensitivity and resource commitment rate.

7. Limitations and Open Questions

5. **Win rate extension:** The win rate component W(t) extends the basic LIF model by adding a second timescale. A formal analysis of the combined P(t) = E(t)×W(t) integral as an extension of the basic LIF model is deferred to future work. The relevant framework is the two-timescale filtering literature.
6. **Stochastic generalization:** The power metric includes Brownian motion terms in its full stochastic formulation (dE = μE dt + σE dW, documented in the patent). The stochastic LIF model (with noise current) is well studied. The correspondence between the two stochastic versions requires formal analysis.
7. **Biological validation:** The LIF model is a simplification of real neural dynamics. More detailed models (Hodgkin-Huxley, conductance-based models) deviate from the pure

exponential decay. Whether the resource commitment principle holds in more realistic neural models is an open question.

8. **Optimality domain:** The optimality proof assumes squared-error loss with exponential forgetting. Different loss functions or memory structures would yield different optimal kernels. The present result is optimal within this class.

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8. Conclusion

We have identified and proved the Resource Commitment Principle: the unique optimal solution to the problem of evidence-based resource allocation under uncertainty is the exponential decay integral with kernel $h(t) = e^{(-\lambda t)}$. This principle is instantiated independently in six fields across three centuries of scientific work: electronics, statistics, neuroscience, cognitive psychology, stochastic finance, and financial risk management.

The structural correspondence between the LIF neuron model and the power metric holds exactly at the level of the integral kernel (numerical error = 0.00). This is an algebraic identity for the discretized integral form — not a full biophysical equivalence. The reset mechanism differs: classical LIF resets after firing; the power metric produces a soft-reset through decay and thresholding. Within that boundary, it is the same differential equation with the same solution, derived independently in two fields separated by a century of scientific development. Biological evolution and financial engineering solved the same optimization problem and produced the same kernel.

For the applied series (Papers 1-4), this result is foundational: the power metric framework applies without modification across ML training, inference compute, scaling law reliability, and data mixing because each of these domains presents the Resource Commitment Problem in a different guise. The framework is not a domain-specific tool. It is a domain-agnostic solution to a universal problem. The fact that biological brains discovered the same solution first is perhaps the strongest possible validation.

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Contact & Collaboration

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This is the theoretical foundation paper for the adaptive compute allocation series. It does not require experimental validation — the mathematical identity is proven analytically and verified numerically. Collaboration welcomed on the stochastic generalization (win rate as two-timescale filter) and the biological validation question. Christopher Ré (Stanford) and computational neuroscience colleagues are the target collaborators for the biological extension.